

Anomaly based detection

1. The research paper focuses on IDS which is working based on PCC – CNN i.e Pearson correlation coefficient for streamlined feature selection with Convolution Neural Network for traffic classification.
2. This approach has a superiority because the accuracy is 99.89 while maintaining a low false alarm rate.
3. This identifies both binary (Normal traffic vs malicious traffic) and multiclass attacks(DDOS, DOS, Probing, R2L and U2R)
4. Findings suggest that PCC – CNN is a lightweight and computationally efficient model.

What are Signature based and rule based?

Signature based are the ones which look for a specific match i.e the cryptographic hash or string or a key that matches a bad looking attack script or so.

Rule based are the ones which works based on the logic that is written by any of the experts and it keeps checking the condition and can validate various data and metadata at the same time and the main backdrop of this method is that it can go for a long amount of time if the logic is too big or the condition to be checked is quite big.

Linear feature extraction is a method where multiple features are compared together using PCC and if the PCC value is near to +1 then it stands that they are highly similar while 0 stands for no relationship and -1 stands for highly negative relationship.

EX : if two features has a PCC ~ 0.85 it means that they are likely similar and one of the feature is discarded.

PCC – CNN model is a state of the art model which is trained on 3 datasets NSL – KDD, CICIDS – 2017 and IOTID20 and compared with 5 other different ML models having an accuracy of 99.98% and a false alarm rate of 0.02%.

PCC is a filter based feature selection technique to identify highly valuable features which are then passe to CNN and is then checked for the binary and multiclass network anomaly classification.

Anomaly based creates a lot of false positives, therefore it is important to update it based on latest ML and AI models.

Commonly used datasets to design IDS are KDDCUP99, DARPA but the problem is that they have been created long ago and even the redundancy of the features between the classes

Weka is a tool that has different types of ML and DL classifiers to predict and compare how algorithms would perform.

1. NSL KDD : This dataset consists of 4 categories and the attacks can fall into any of them and they are DoS, R2L, U2R, Probe.
2. CICIDS2017 : This dataset consists of 4 categories and there are 11 types of simulated attacks that fall into these 4 categories and they are BENIGN, DoS, Firewall, Port Scan.
3. IOTID20 : uses raw packet files and this dataset has been purely created for academic purposes and this one has been categorized into 4 types Dos, Mirai, Scan, MITM ARP.

PIPELINE

It consists of

1. Data preprocessing : data is cleaned and all the unwanted features are removed and then the Null values /NaN values are replaced or deleted.
2. Feature Selection : It is very important as it increases the accuracy of the model and even prevents overfitting of the models and reduces the cost and training time of the model too, and this one does feature selection works on pearson correlation coefficient method and this method calculates the coefficient by evaluating the correlation between the instance and a class and to reduce overfitting we select the features contributing most to classification.
3. ML Algorithm : This consists of various ML algorithms that have been implemented and they are
 - a. Logistic Regression : This estimates the probability of a new event being a malicious one based on the previous events.
 - b. LDA : predicts the likelihood that a new set belongs to every class.
 - c. KNN : estimates how likely a point is to be a member of one or more groups
 - d. CART
 - e. SVM
4. PCC – CNN : This model builds an efficient IDS and this is a lightweight process because it will keep only the required data while discarding the whole unnecessary data. PCC is a filter based algorithm as well as feature selection technique. It even uses ReLU which is Rectified Linear Unit and this is an activation function and outputs the input directly if it is positive i.e if $x \geq 0$ then x is the output directly else it outputs 0.
5. Evaluation : This has been evaluated using Confusion matrix, Accuracy, Precision, F1-Score, False Alarm Rate, ROC (Receiver Operating Characteristics) – AUC(Area under Curve) Curve.

ALTERNATIVE DATASETS

1. UNSW-NB15 : This dataset consists of network traffic dataset that is simulating the real attack environments and is widely used along with NSL-KDD and this is considered more realistic than KDD Cup 99 for IDS evaluation and this is created using the IXIA perfect storm tool in a cyber range environment combining normal modern activities with contemporary attack behaviors.
 - a. One of the cons of the dataset is that some attack categories are very small such as worms and shellcode and because of this, the model can obtain a high overall accuracy while performing poorly on minority attack classes.
2. BoT-IoT : This contains IoT specific botnet traffic from the **CYBER RANGE LAB** at UNSW Canberra used in ensemble and deep learning IDS which achieved up to 99.6% accuracy.
 - a. The biggest cons of this dataset is the class imbalance and this contains an enormous amount of attack traffic compared to normal traffic and because of this it can achieve extremely high accuracy but it can be a poor IDS.
3. TON_IoT : captures the telemetry from IoT Devices and has been used with hybrid CNN-LSTM models reaching 98.75% binary classification.
 - a. One of the major disadvantage of this data set is that it is more complicated because there are a lot of data source and the feature selection becomes hard and the feature fusion becomes tough too.
4. CICIoT2023 : real world IoT traffic from 105 devices with 33 attack types across 7 categories achieving upto 99.995% binary accuracy with DL.
 - a. The major problem in this dataset is again the class imbalance and the issue is that 33 individual attacks are much harder to classify than 7 attack groups.
5. CICDDoS2019 : this focuses mainly on DDoS attack scenarios and has been evaluated with multiple ML algorithms reaching above 99% accuracy.
 - a. It is good for binary classification and is good for multiclass classification too but the issue is that almost each and every attack belongs to the same family i.e **DDoS** which can lead the model towards learning what is the current type of DDoS attack rather than finding what type of cyber attack is happening.
 - b. This doesn't provide the broad attack diversity of UNSW-NB15 or CICIoT2023

Purpose	Best dataset
General IDS benchmark	UNSW-NB15
IoT binary IDS	CICIoT2023
IoT multiclass IDS	CICIoT2023
Fine-grained IoT attack classification	CICIoT2023
IoT/IIoT heterogeneous IDS	TON_IoT
Botnet-focused IDS	BoT-IoT
DDoS-specific IDS	CIC-DDoS2019

CICIoT2023 is the best alternative or an additional dataset as the IDS is fledged for IoT devices while UNSW – NB15 is the cleaner general purpose benchmark and both of these can be used to work on both binary and multiclass

1. BOT-IoT

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- b. <https://www.sciencedirect.com/science/article/abs/pii/S2542660523001427?via%3Dihub>
- c. <https://www.mdpi.com/2076-3417/15/4/2043>
- d. <https://www.nature.com/articles/s41598-025-85248-z>
- e. <https://ijeecs.iaescore.com/index.php/IJECS/article/view/32573>
- f.

2. UNSW – NB 15

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- c. <https://www.nature.com/articles/s41598-025-95621-7>
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- e. <https://ieeexplore.ieee.org/document/10989358>
- f. <https://ieeexplore.ieee.org/document/10820504>
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3. TON – IoT

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- b. <https://www.sciencedirect.com/science/article/abs/pii/S2542660523001427?via%3Dihub>
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4. CICIoT2023

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5. CICDDoS2019

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- c. <https://ieeexplore.ieee.org/abstract/document/10730435>
- d. https://link.springer.com/chapter/10.1007/978-981-19-2948-9_38
- e. <https://etasr.com/index.php/ETASR/article/view/14119>
- f. <https://ieeexplore.ieee.org/abstract/document/11409660>
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6. NSL KDD

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- b. <https://ieeexplore.ieee.org/document/8740962>
- c. <https://www.sciencedirect.com/science/article/pii/S1877050924006458?via%3Dihub>

- d. <https://ijiesia.com/ijiesia/article/view/19>
- e. <https://www.nature.com/articles/s41598-026-59610-8>
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7. CICIDS2017

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